Joseph Breda

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Research

My impact-driven research focuses on developing and evaluating **AI** alongside **population-scale signals** to enhance **societal health and well-being**. I have designed and led large-scale evaluations of LLMs for differential diagnosis via population deployments, directed a multi-year flu challenge study with wearables to use digital biomarkers to predict pre-symptomatic onset of the flu, and built geospatial computer vision models to map traffic safety. My work addresses the practical challenges of aligning society-scale technologies with human-centric applications and real populations.

Education

2019 - 2026 Ph.D. — **University of Washington,** *Computer Science*
Thesis: *Ubiquitous Computing Platforms as Scalable Surrogates in Public Health and Well-being (Defended 2025)*
Advisor: *Shwetak Patel*

2015 - 2019 B.S. — **University of Massachusetts Amherst**, *Electrical & Computer Engineering*
Advisor: *Jay Taneja*

Select Projects

SymptomAI: Conversational AI Agents for Differential Diagnosis — *Nature, In Submission*

- Leading population-scale deployment and clinical evaluation of LLMs for differential diagnosis with real patients (N=13,000), validating LLM verifier for scalable oversight.
- Bridging validated AI diagnoses with population-scale wearable data, enabling association of digital biomarkers with disease phenotypes at scale.

Biomarkers for Flu Onset Prediction via Clinical Challenge Study — *Nature, In Preparation*

- Leading flu challenge study in collaboration with NIH/NIAID, deploying commodity wearables with patients intentionally infected with the H10N7 “bird” flu.
- Modeled pre-symptomatic immune activation with wearable biomarkers, predicting flu onset, validated against gene expression from longitudinal bloodwork.

FeverPhone: Accessible Fever Monitoring with Unmodified Smartphones — *IMWUT, 2023*

- Developed ML model mapping smartphone temperature during user-contact to measure core-body temperature, evaluated in clinic with real patients.
- Won ***Distinguished Paper Award*** at Ubicomp 2024, Highlighted in GetMobile Magazine.

ProxiCycle: Passively Mapping Cyclist Safety with Near-Miss Detection — *CHI, 2025*

- Developed custom embedded system for passively crowdsensing geospatial cyclist safety, validated through 2-month longitudinal deployment with 15 cyclists (1000 miles traveled).
- Received national media attention, featured on NPR All Things Considered.

Double-Blind Deployment Study of LLMs for Triaging in Emergency Room — *Working Paper*

- Designed and ran prospective double-blind study deploying LLM chatbots in clinical waiting rooms to capture diagnostic reasoning and compare it against physician notes.
- Building mixed-methods analysis pipeline to characterize LLM failure modes and identify systematic divergence between LLM and clinical-expert reasoning turn-by-turn.

Industry Experience

Google Student Researcher *July 2025-Present*

- Leading SymptomAI project, evaluating conversational AI for differential diagnosis in population-scale deployment (N=13,000).
- Fine-tuning and evaluating wearable foundation model for downstream women's health forecasting tasks, targeting improvements on underrepresented data.

Google Student Researcher *May 2021-Oct. 2022*

- Developed geospatial deep learning model forecasting collision rates from visual road features and historic user data.

Google SWE Intern *May 2020-Sept. 2020*

- Developed synthetic population generation pipeline used for evaluation of Google Maps traffic models and epidemiological models of disease spread during Covid-19 pandemic.

Publications

P19 *SymptomAI: Towards a Conversational AI Agent for Everyday Symptom Assessment*
Joseph Breda, Fadi Yousif, Beszel Hawkins, Marinela Cotoi, Miao Liu, Ray Luo, Po-Hsuan Cameron Chen, Mike Schaekermann, Samuel Schmidgall, Xin Liu, Girish Narayanswamy, Samuel Solomon, Maxwell A. Xu, Xiaoran Fan, Longfei Shangguan, Anran Wang, Bhavna Daryani, Buddy Herkenham, Cara Tan, Mark Malhotra, Shwetak Patel, John B. Hernandez, Quang Duong, Yun Liu, Zach Wasson, Dimitrios Antos, Bob Lou, Matthew Thompson, Jonathan Richina, Anupam Pathak, Nichole Young-Lin, Jake Sunshine, Daniel McDuff
Arxiv 2026

P18 *FlowRing: Integrated Microgesture and Surface Interaction for Versatile XR Input*
Ishan Chatterjee, Jiexin Ding, Anandghan Waghmare, **Joseph Breda**, Yuquan Deng, Bo Liu, Yuntao Wang, Shwetak Patel
MHCI 2025

P17 *Lessons Learned From FeverPhone: Towards Scalable, Accessible At-Home Diagnostics via Fever Detection on Unmodified Smartphones*
Joseph Breda, Mastafa Springston, Alex Mariakakis, Shwetak Patel
GetMobile 2025

P16 *MobileSense: Tracking Air Pollution Exposure of Frontline Motorcycle Riders Using a Mobile Low-Cost Sensor Platform*
Raunak Mukhia, Adisorn Lertsinsrubtavee, Preechai Mekbungwan, Kalana GS Jayarathna, Nussara Tieanklin, **Joseph Breda**, Ratchaphon Samphutthanont, Thongchai Kanabkaew, Kurtis Heimerl, Kanchana Kanchanasut
COMPASS 2025

P15 *The Air We Share: Seven-Month Mixed Method Study of Thailand's Motorcycle Rideshare Driver and Air Pollution*
Tieanklin, Chanwut Kittivorawong, **Joseph Breda**, Adisorn Lertsinsrubtavee, Preechai Mekbungwan, Raunak Mukia, Kalana G.S. Jayarathna, Ratchaphon Samphutthanont, Chayanon Sawatdeenarunat, Kurtis Heimerl
Computing and Sustainable Societies 2025

P14 *ProxiCycle: Passively Mapping Cyclist Safety Using Smart Handlebars for Near-Miss Detection*
Joseph Breda, Keyu Chen, Thomas Ploetz, Shwetak Patel
CHI 2025

P13 *NightLight: Passively Mapping Nighttime Sidewalk Light Data for Improved Pedestrian Routing*
Joseph Breda*, Daniel Campos Zamora*, Shwetak Patel, Jon Froehlich
CHI 2025

P12 *Exploring and Characterizing Large Language Models for Embedded System Development and Debugging*
Zachary Englhardt, Richard Li, Dilini Nissanka, Zhihan Zhang, Girish Narayanswamy, **Joseph Breda**, Xin Liu, Shwetak Patel, Vikram Iyer
CHI Late Breaking Work 2024

P11 *'I will just have to keep driving': A Mixed-methods Investigation of Lack of Agency within the Thai Motorcycle Rideshare Driver Community*
*Nussara Tieanklin, ***Joseph Breda**, Tim Althoff, Kurtis Heimerl
CSCW 2024

P10 *Thermal Earring: Low-power Wireless Earring for Longitudinal Earlobe Temperature*
Qiuyue Shirley Xue, Yujia Liu, **Joseph Breda**, Mastafa Springston, Vikram Iyer, Shwetak Patel
IMWUT 2024

P9 *Understanding People's Concerns and Attitudes Toward Smart Cities*
Pardis Emami-Naeini, **Joseph Breda**, Wei Dai, Tadayoshi Kohno, Kim Laine, Shwetak Patel, Franziska Roesner
CHI 2024

P8 🏆 *Feverphone: Accessible Core-Body Temperature Sensing for Fever Monitoring Using Commodity Smartphones*
Joseph Breda, Mastafa Springston, Alex Mariakakis, Shwetak Patel
IMWUT 2023 **Won Distinguished Paper Award**

P7 *SpiroMask: Measuring Lung Function Using Consumer-Grade Masks*
Rishiraj Adhikary, Dhruvi Lodhavia, Chris Francis, Rohit Patil, Tanmay Srivastava, Prerna Khanna, Nipun Batra, **Joseph Breda**, Jacob Peplinski, Shwetak Patel
ACM Transactions on Computing for Health 2023

P6 *Passively Sensing SARS-CoV-2 RNA in Public Transit Buses*
Jason Hoffman, Matthew Hirano, Nuttada Panpradist, **Joseph Breda**, Parker Ruth, Yuanyi Xu, Jonathan Lester, Bichlien H. Nguyen, Luis Ceze, Shwetak Patel
Science of the Total Environment 2022

P5 *Phone-based Ambient Temperature Sensing Using Opportunistic Crowdsensing and Machine Learning*
Amee Trivedi, Phuthipong Bovornkeeratiroj, **Joseph Breda**, Prashant Shenoy, Jay Taneja
Sustainable Computing 2021

P4 *Hanging Gardens of Babylon: Reframing Urban Agriculture as an Opportunity for Social Engagement*
Joseph Breda, Esther Jang, Kurtis Heimerl, Shwetak Patel
Self-Sustainable CHI 2020

P3 *Hot or Not: Leveraging Mobile Devices for Ubiquitous Temperature Sensing.*
Joseph Breda, Amee Trivedi, Chulabhaya Wijesundara, Phuthipong Bovornkeeratiroj, David Irwin, Prashant Shenoy, Jay Taneja
BuildSys 2019

P2 *Staring at the Sun: A Physical Black-box Solar Performance Model*
Dong Chen, **Joseph Breda**, David Irwin
BuildSys 2018

P1 *Fancy That: Measuring Electricity Grid Voltage Using a Phone and a Fan.*
Joseph Breda and Jay Taneja
COMPASS 2018

Organizing Experience

O2 **Founded and President of the Allen School Graduate Entrepreneurship Club** — 2023-Present
Organizing quarterly panels and event connecting academic start up founders with current PhD students interested entrepreneurship.

O1 **Founding member of CS4Env cross-department collaborative initiative.** — 2022-2023
Assisted with early organizing and developed website during first year launch.

Teaching Experience

T3 **Embedded Systems Capstone Teaching Assistant** — Winter 2024
See [T2].

T2 **Embedded Systems Capstone Teaching Assistant** — Spring 2024
Mentored teams of students on end-to-end capstone projects and led lectures on embedded ML and Android BLE.

T1 **Embedded Systems Teaching Assistant** — Fall 2023
Tutored embedded systems during office hours and graded assignments

Academic Service

AS7 **UW CSE PhD Area Chair** — 2025
Triaged over 30 PhD Applications and served as area chair in support of Ubicomp and HCI faculty decisions on PhD Admissions

AS6 **Paper Reviewer** — 2025
Reviewed for ACM IMWUT, ACM CHI, ACM UIST & ACM COMPASS

AS5 **Paper Reviewer** — 2024
Reviewed for ACM IMWUT & ACM CHI

AS4 **High school Intern Outreach and Mentor** — 2024
Interviewed and mentored 1 team of 5 high school summer lab interns who built a cognitive testing game.

AS3 **Paper Reviewer** 2023
Reviewed for ACM CSCW

AS2 **High school Intern Outreach and Mentor** 2023
Interviewed and mentored 2 teams of 4 high school summer lab interns who built a zoom-call AI summarization platform.

AS1 **UW CSE PhD Admission Reader** 2022
Triaged over 60 PhD Applications

Awards

A10 **Samsung Global Research Outreach (GRO) Grant** November 2024
Primary author on **$150,000** grant proposal extending my prior work collaborating with the NIH for flu monitoring and early detection.

A9 **Distinguished Paper Award at Ubicomp 2024** October 2024
For work on my first author paper [P8].

A8 **AI-Focused Population Health Pilot Grant** June 2024
Primary contributor to **$100,000** university level pilot grant to for population health applications of AI

A7 **Young Researcher** October 2023
Invited and attended 10th Heidelberg Laureate Forum

A6 **Computing for the Environment Initiative Grant** June 2022
Primary author on 2 project proposals totaling **$100,000** of funding ($50,000 each) for developing computer systems for sustainability.

A5 **Google Gift Grant** October 2021
Primary author of **$60,000** grant to study human mobility patterns.

A4 **Weil Family Endowed Fellowship in Computer Science & Engineering** September 2019
Selected for award upon PhD admission.

A3 **Graduated from Commonwealth Honors College** May 2019
For completing honors undergraduate thesis, later published as [P3].

A2 **Graduated Magna Cum Laude** May 2019
Top 10% of graduating class within the ECE department.

A1 **Commonwealth Honors College: Honors Research Grant** December 2018
Awarded research funding for proposed honors thesis.

Select Press

PR3 **NPR All Things Considered:** This week in science: biker safety, orange cats and a gum disease-heart rhythm link 2025

PR2 **GeekWire:** UW develops bike-mounted sensor system to track close-passing cars and map safer cycling routes 2025

PR1 **Gizmodo**: Researchers Created a Simple App That Turns Your Smartphone Screen Into an Accurate Thermometer 2023

Select Skills

Machine Learning & AI: AI Evaluation Frameworks | LLM-as-Judge / Model Verifiers | Foundation Model Fine-Tuning | Multimodal Learning | Computer Vision | Time-Series Modeling | Signal Processing | Geospatial ML | Deep Learning | Failure Mode Analysis | Human-AI Interaction Studies

Research Methodology: Empirical Analysis | Experimental Design | Ablation & Robustness Testing | Randomized Prospective Study Design | Clinical Validation | Expert Annotation Study | Population Deployment | Pre-Registration & IRB Protocols | Quantitative & Qualitative Methods | Statistical Analysis & Causal Inference

Languages & Frameworks: Python (Primary) | Pandas | Numpy | PyTorch | Tensorflow | Scikit-Learn | C / C++ | Android Development

Engineering: End-to-End ML Pipelines | Multi-Agent Systems | ETL Pipelines | System Scaling | Embedded Systems | Hardware Prototyping | Computational Fabrication | Real-World Data Pipelines

Communication & Collaboration: Pair Programming | Scientific Writing & Public Information Dissemination | Cross-Disciplinary Collaboration | Translational Research | Mentorship | Grant Writing & Project Proposal